

EXPERIMENT (8)

AIM

To perform Sub Query and Correlated Query on the Hostel Management System tables.

DESCRIPTION

A subquery is a query nested inside another SQL query. A correlated subquery depends on the outer query and is evaluated for each row.

SYNTAX

SUB QUERY

```
SELECT column_name  
FROM table_name  
WHERE column_name IN  
(SELECT column_name FROM table_name WHERE condition);
```

CORRELATED QUERY

```
SELECT *  
FROM table_name t1  
WHERE EXISTS  
(SELECT * FROM table_name t2 WHERE t1.column_name=t2.column_name);
```

Questions and Answers

1. Display the rooms whose monthly rent is greater than the average monthly rent.

```
SELECT *  
FROM Hostel_Rooms  
WHERE RentPerMonth >  
(  
SELECT AVG(RentPerMonth)  
FROM Hostel_Rooms  
);
```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

130 FROM Hostel_Warden
131 ORDER BY Experience DESC;
132 • SELECT *
133 FROM Hostel_Rooms
134 WHERE RentPerMonth >
135 (
136     SELECT AVG(RentPerMonth)
137     FROM Hostel_Rooms
138 );

```

Result Grid

RoomID	RoomNumber	BlockName	FloorNo	RoomType	Capacity	OccupiedBeds	RentPerMonth
103	B201	B Block	2	Triple	3	2	5500.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Form Editor
Field Types

t 514 Result 515 Result 516 Hostel_Student 517 Hostel_Warden 518 Hostel_Rooms5 Apply

2. Display the rooms having more than one student.

```

SELECT RoomID,
COUNT(StudentID) AS Total_Students
FROM Hostel_Student
GROUP BY RoomID
HAVING COUNT(StudentID)>1;

```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

135 (
136     SELECT AVG(RentPerMonth)
137     FROM Hostel_Rooms
138 );
139 • SELECT RoomID,
140     COUNT(StudentID) AS Total_Students
141 FROM Hostel_Student
142 GROUP BY RoomID
143 HAVING COUNT(StudentID) > 1;

```

Result Grid

RoomID	Total_Students
--------	----------------

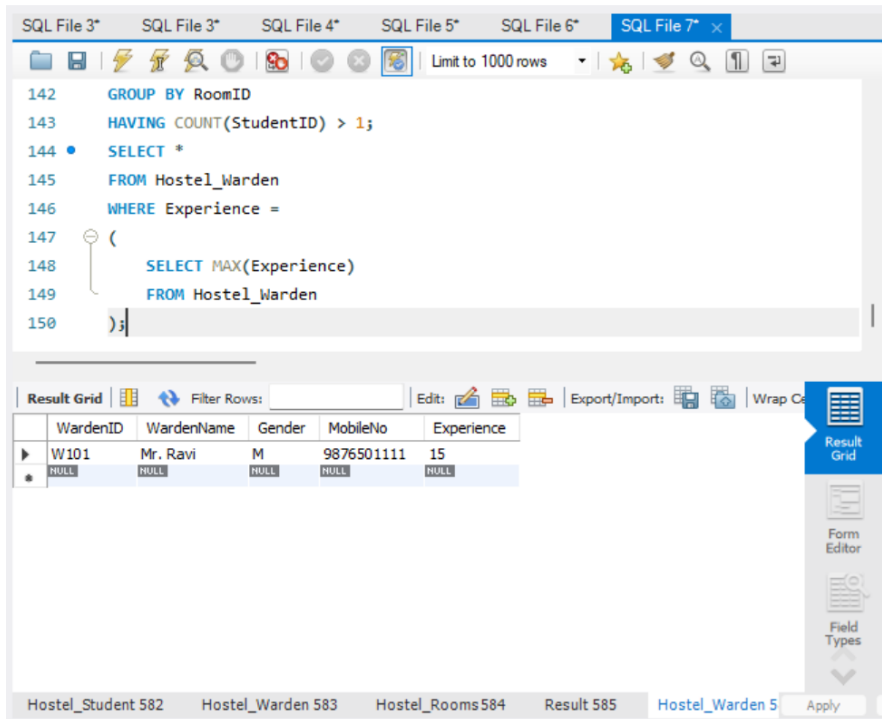
Form Editor
Field Types

t 547 Result 548 Hostel_Student 549 Hostel_Warden 550 Hostel_Rooms551 Result 551 Read Only

Output

3. Display the most experienced hostel warden.

```
SELECT *  
FROM Hostel_Warden  
WHERE Experience=  
(  
SELECT MAX(Experience)  
FROM Hostel_Warden  
);
```



The screenshot shows a SQL IDE interface with a query editor and a result grid. The query editor contains the following SQL code:

```
142 GROUP BY RoomID  
143 HAVING COUNT(StudentID) > 1;  
144 • SELECT *  
145 FROM Hostel_Warden  
146 WHERE Experience =  
147 (  
148 SELECT MAX(Experience)  
149 FROM Hostel_Warden  
150 );
```

The result grid displays the following data:

WardenID	WardenName	Gender	MobileNo	Experience
W101	Mr. Ravi	M	9876501111	15
*	NULL	NULL	NULL	NULL

The IDE interface includes a toolbar with various icons, a status bar at the bottom showing file names (Hostel_Student 582, Hostel_Warden 583, Hostel_Rooms 584, Result 585, Hostel_Warden 5), and a sidebar with options like Result Grid, Form Editor, and Field Types.

4. Display the room details whose rent is greater than the average rent of the same room type.

```
SELECT *  
FROM Hostel_Rooms R1  
WHERE RentPerMonth >  
(  
SELECT AVG(RentPerMonth)  
FROM Hostel_Rooms R2  
WHERE R1.RoomType=R2.RoomType  
);
```

SQL File 3* SQL File 3* SQL File 4* SQL File 5* SQL File 6* SQL File 7* x

Limit to 1000 rows

```

157 (107,'B203','B Block',2,'Double',2,1,5500.00,'Available','Mrs. Priya');
158 • SELECT *
159 FROM Hostel_Rooms R1
160 WHERE RentPerMonth >
161 (
162     SELECT AVG(RentPerMonth)
163     FROM Hostel_Rooms R2
164     WHERE R1.RoomType = R2.RoomType
165 );

```

Result Grid

	RoomID	RoomNumber	BlockName	FloorNo	RoomType	Capacity	OccupiedBeds	RentPerMonth
▶	105	A105	A Block	1	Single	1	1	4500.00
	107	B203	B Block	2	Double	2	1	5500.00
*	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

Form Editor

Field Types

Hostel_Warden 686 Hostel_Rooms687 Result 688 Hostel_Warden 689 Hostel_Rooms689 Apply

RESULT

Thus, the records were successfully retrieved from the Hostel Management System tables using Sub Query and Correlated Query.